Physical Chemistry of Materials

Lecture 16

2차 중간고사 안내 및 복습

2차 중간고사 안내

- 시간: 11월 12일(목) 오후 1:30~2:30
- 장소: 화학관 414호
- 신분증 지참(모바일 학생증도 가능)
- 계산기 불필요

시험 범위

고체(10강)

결정 진동: 고전적인 그림(11강)

결정 진동: 포논(12강)

고체의 전자 구조(13강)

자성(14강)

고체화학의 응용(15강)

시험 문제

총 여섯 문제

개념을 묻는 문제: 세 문제

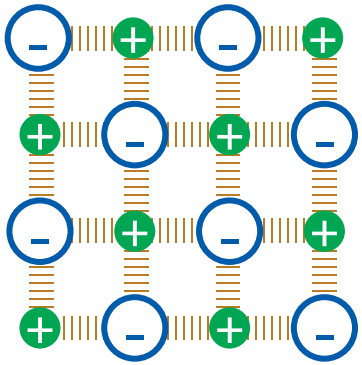
공식을 유도하는 문제: 세 문제

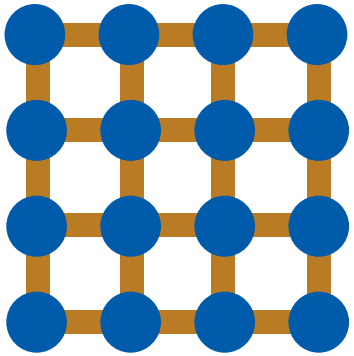
소재물리화학의 구성

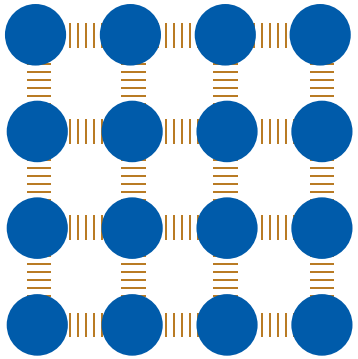
고전 열역학	고체	연성물질
열역학 복습	고체의 구조	액체
상 전이	고체의 진동	유리
통계역학	고체의 전기적 성질	고분자의 구조
기초 이론	고체의 자기적 성질	고분자 용액
이상 기체	표면화학	고분자의 점탄성
상호작용 계	반도체, 나노화학, …	생체 고분자

Bonding in Solids

- Ionic solids: ionic bonds
- Covalent solids: covalent bonds
- Molecular solids: intermolecular interactions





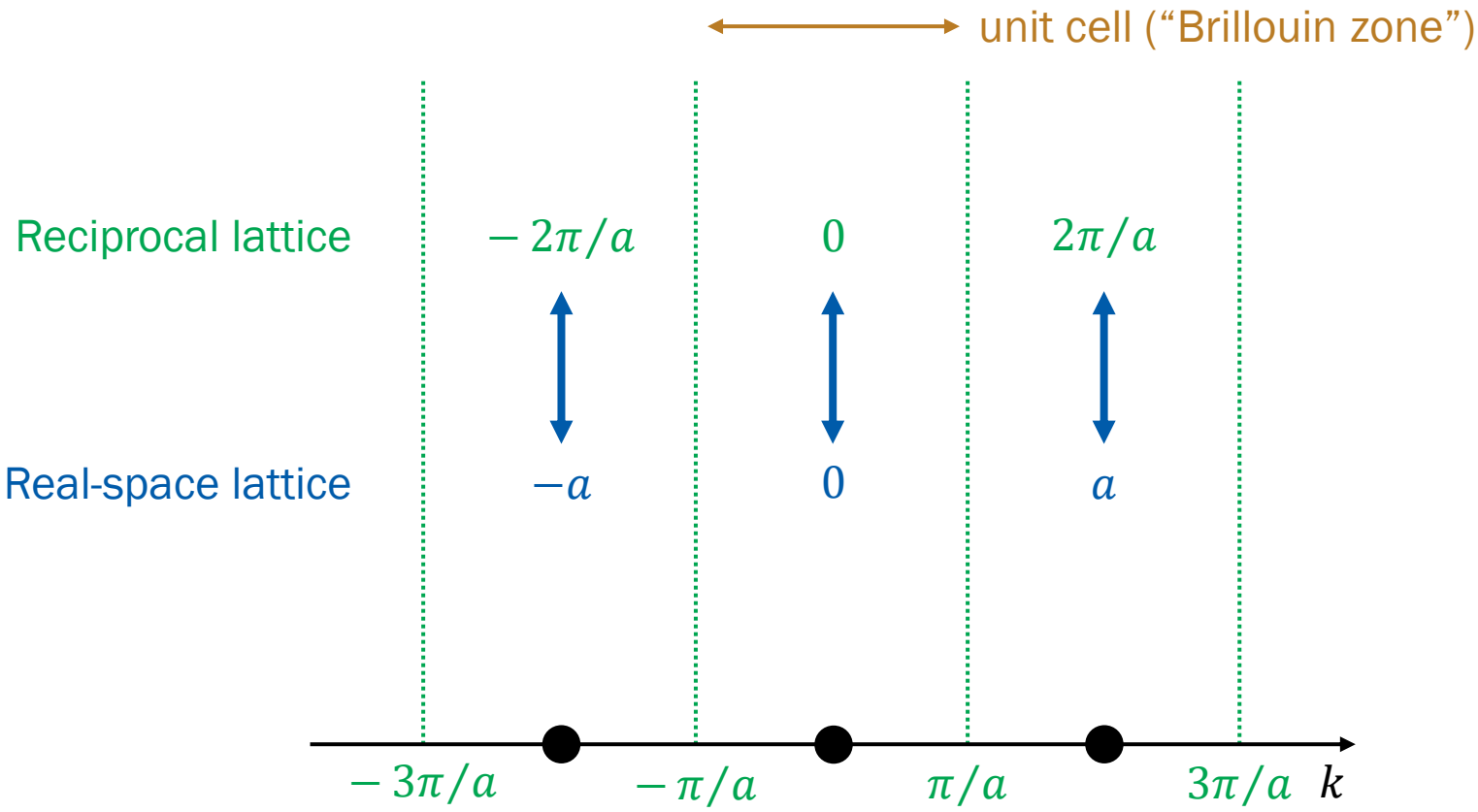


Metals need more discussion

Structures of Solids

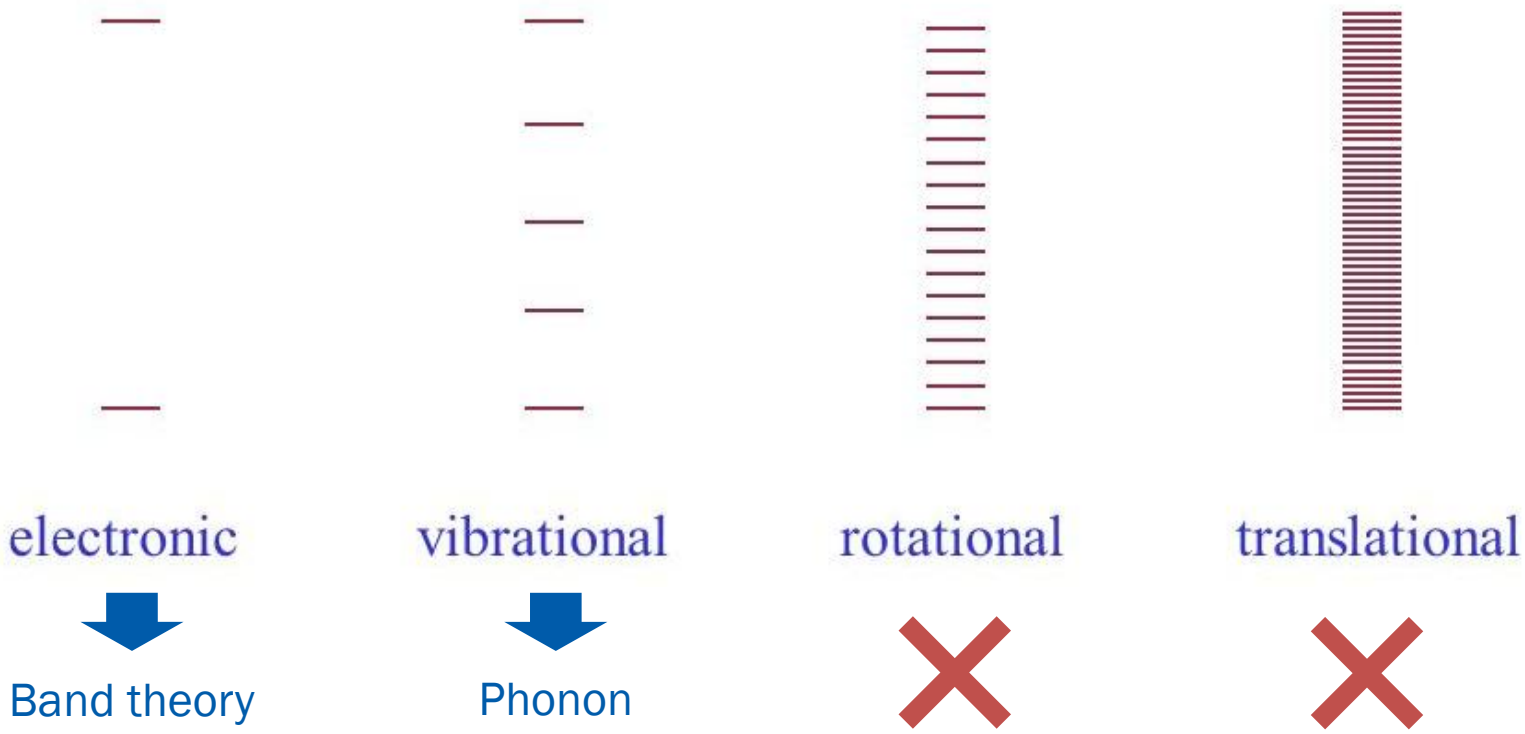
- Unit cell
- Primitive unit cell
- Lattice constants
- Lattice systems, Bravais lattices
- Crystal symmetry
- Close packing: cubic (CCP) and hexagonal (HCP)
- Quasicrystals

Reciprocal Lattice



Molecules and Solids

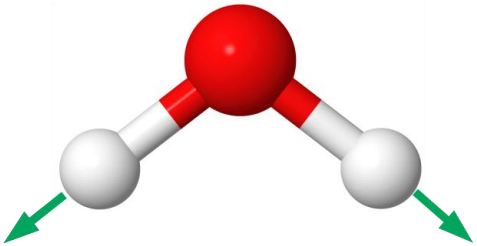
Molecular energies:



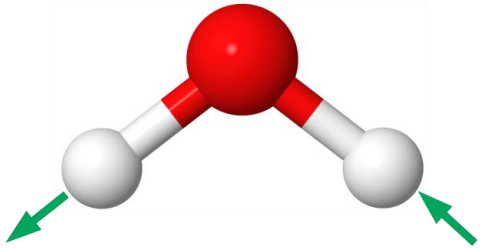
(Image: https://slideplayer.com/slide/6282680/)

Normal Modes

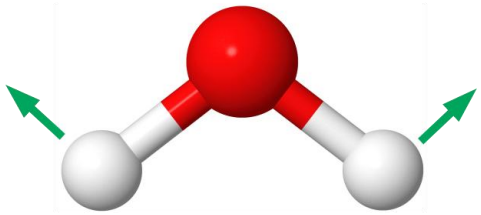
a pattern of motion in which all atoms move with the same frequency



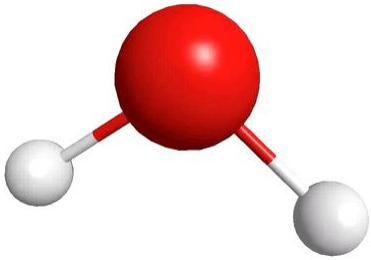
symmetric stretching



asymmetric stretching

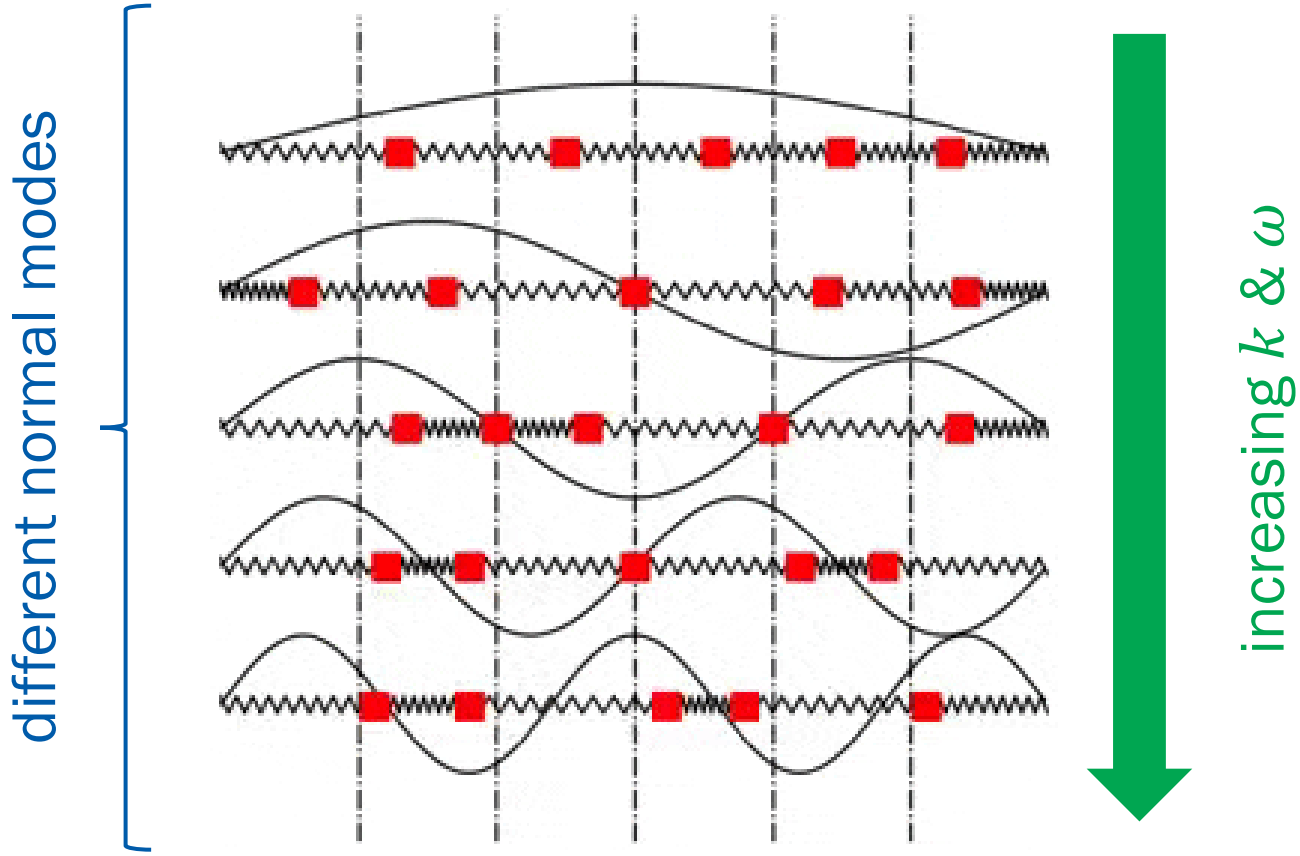


bending



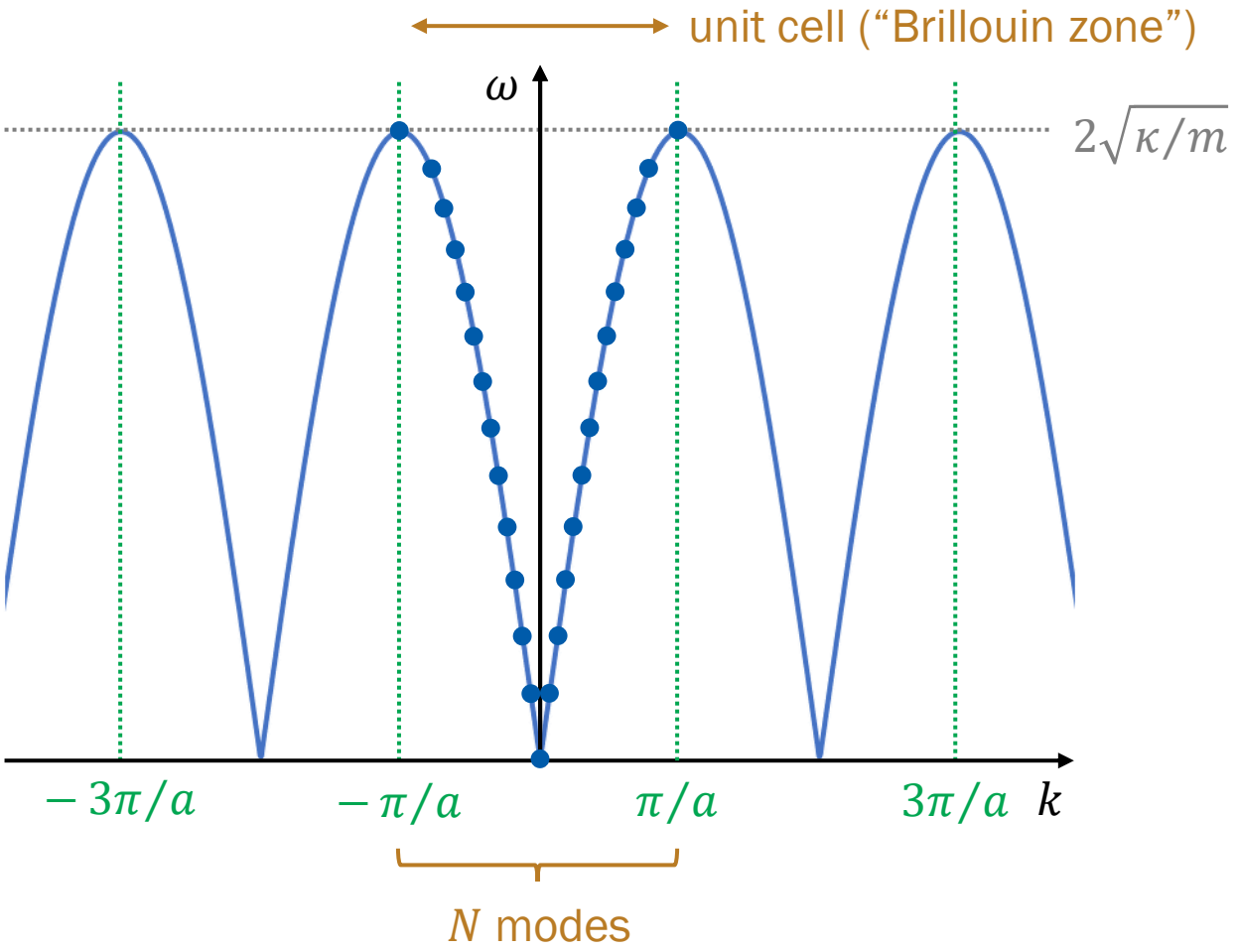
(Images: https://strawberryfields.ai/photonics/_images/water.png
https://thepaperccny.com/2022/02/28/harmony-of-the-ocean/)

Lattice Normal Modes (N = 5)

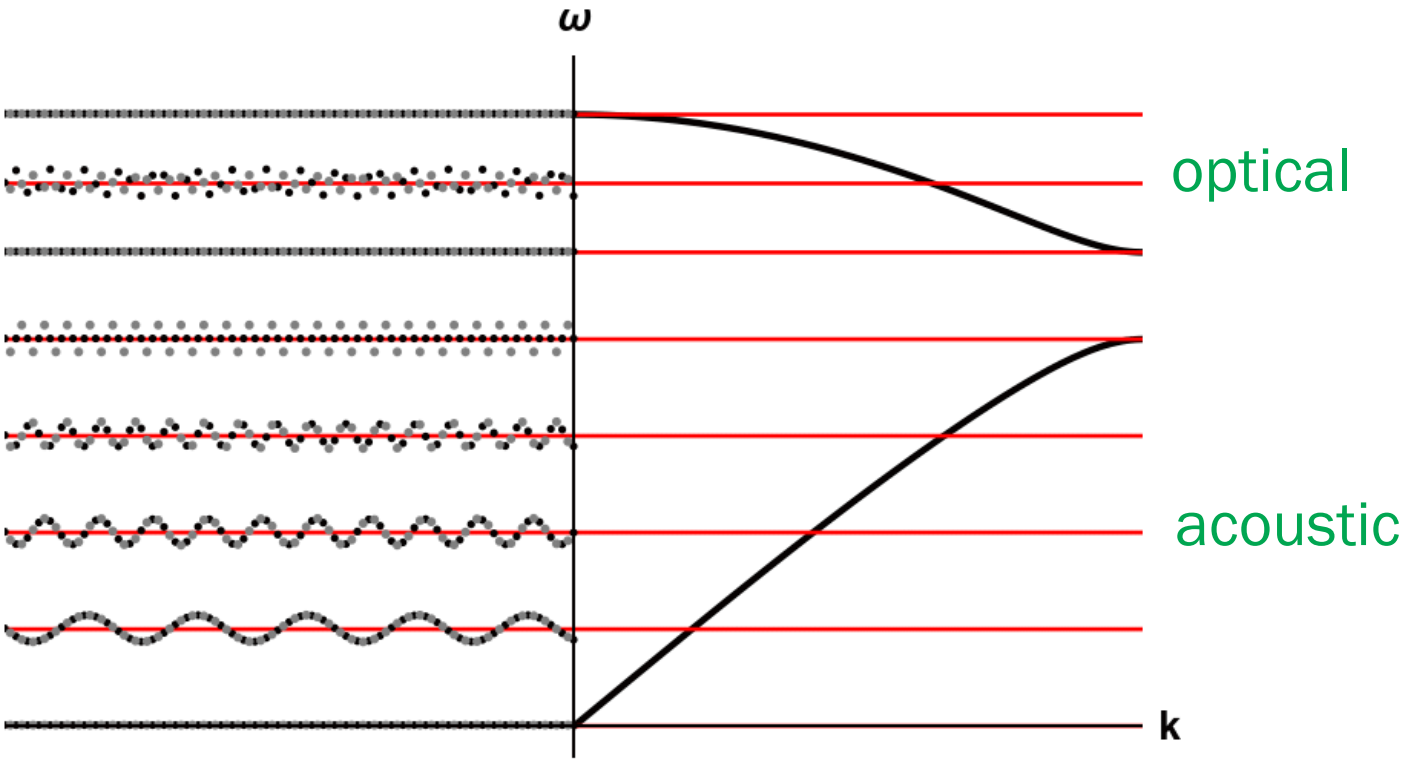


(Image: https://www.quora.com/What-is-an-intuitive-explanation-of-phonons)

For Large N


unit cell (“Brillouin zone”)
ω
$2\sqrt{\kappa/m}$
$-3\pi/a$
$-\pi/a$
π/a
$3\pi/a$
k
N modes

Diatomic Solids



(Image: https://upload.wikimedia.org/wikipedia/commons/d/d7/Diatomic_chain.gif)

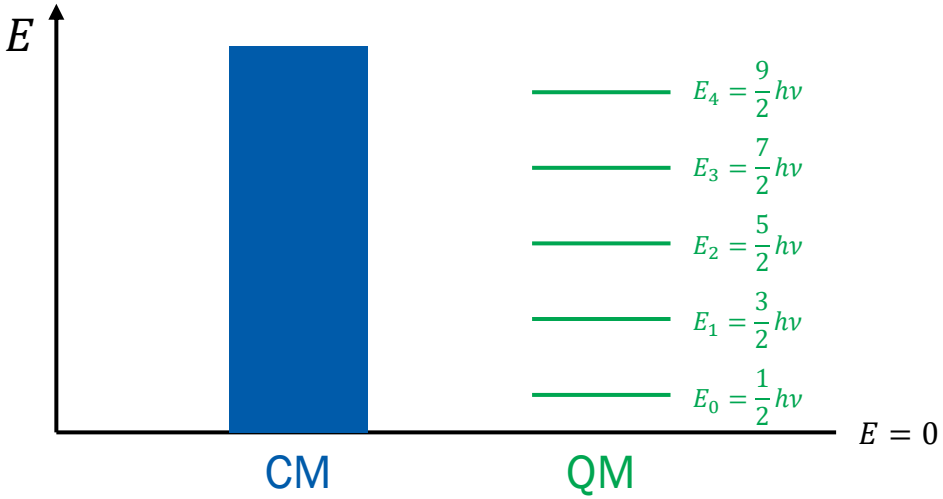
Stat Mech of Crystal Vibration

	Identical oscillators	Normal modes
Classical	Dulong-Petit	Dulong-Petit
Quantum	Einstein model	Debye model

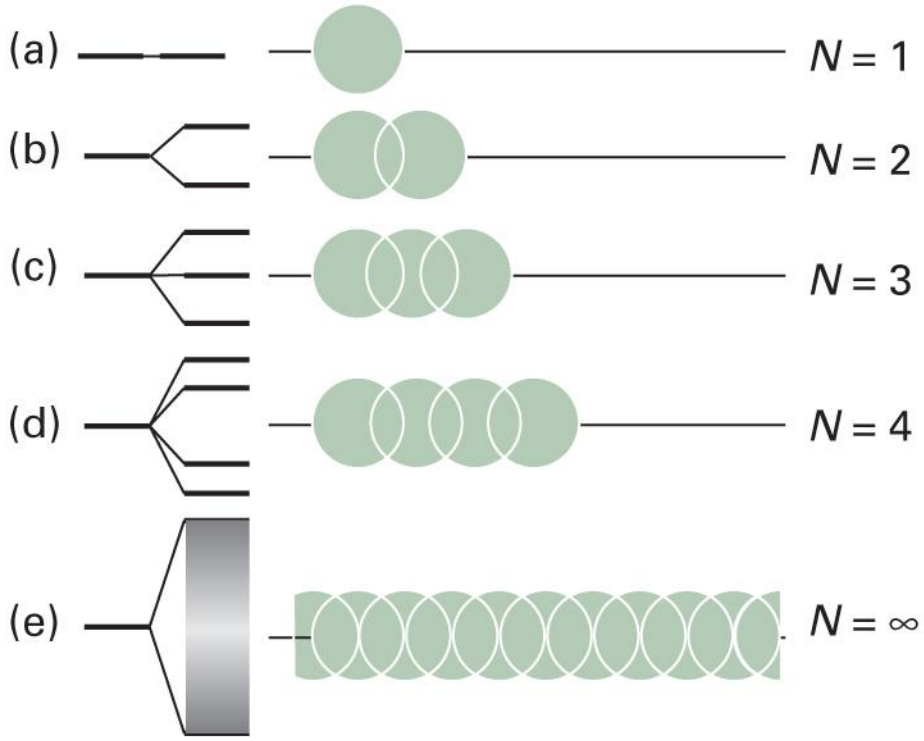
“Phonon”

A phonon is a discrete quantum of vibration

(bosonic)



Electronic Structures of Solids

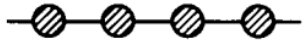


(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 11th ed., Figure 15C.6)

Bands

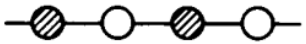
$$\psi_0 = \sum_n e^0 \chi_n = \sum_n \chi_n$$

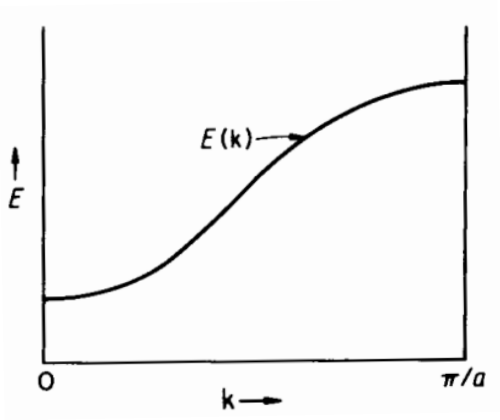
$$= \chi_0 + \chi_1 + \chi_2 + \chi_3 + \cdots$$



$$\psi_{\frac{\pi}{a}} = \sum_n e^{\pi i n} \chi_n = \sum_n (-1)^n \chi_n$$

$$= \chi_0 - \chi_1 + \chi_2 - \chi_3 + \cdots$$



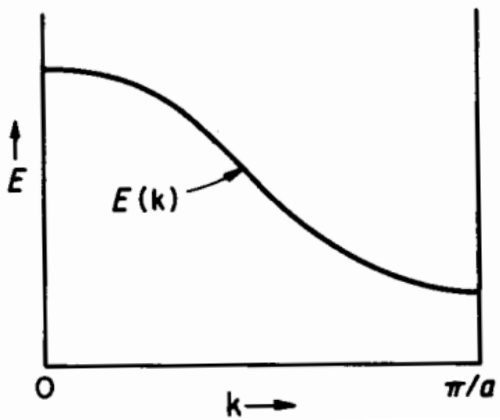


$$\psi_0 = \chi_0 + \chi_1 + \chi_2 + \chi_3 + \cdots$$



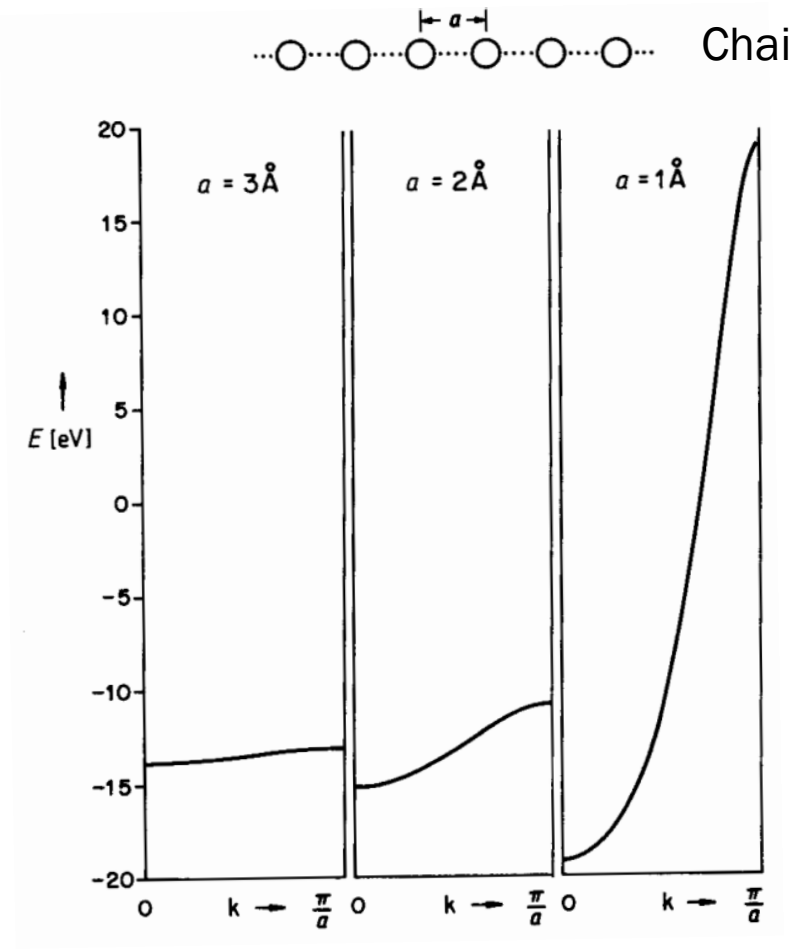
$$\psi_{\frac{\pi}{a}} = \chi_0 - \chi_1 + \chi_2 - \chi_3 + \cdots$$





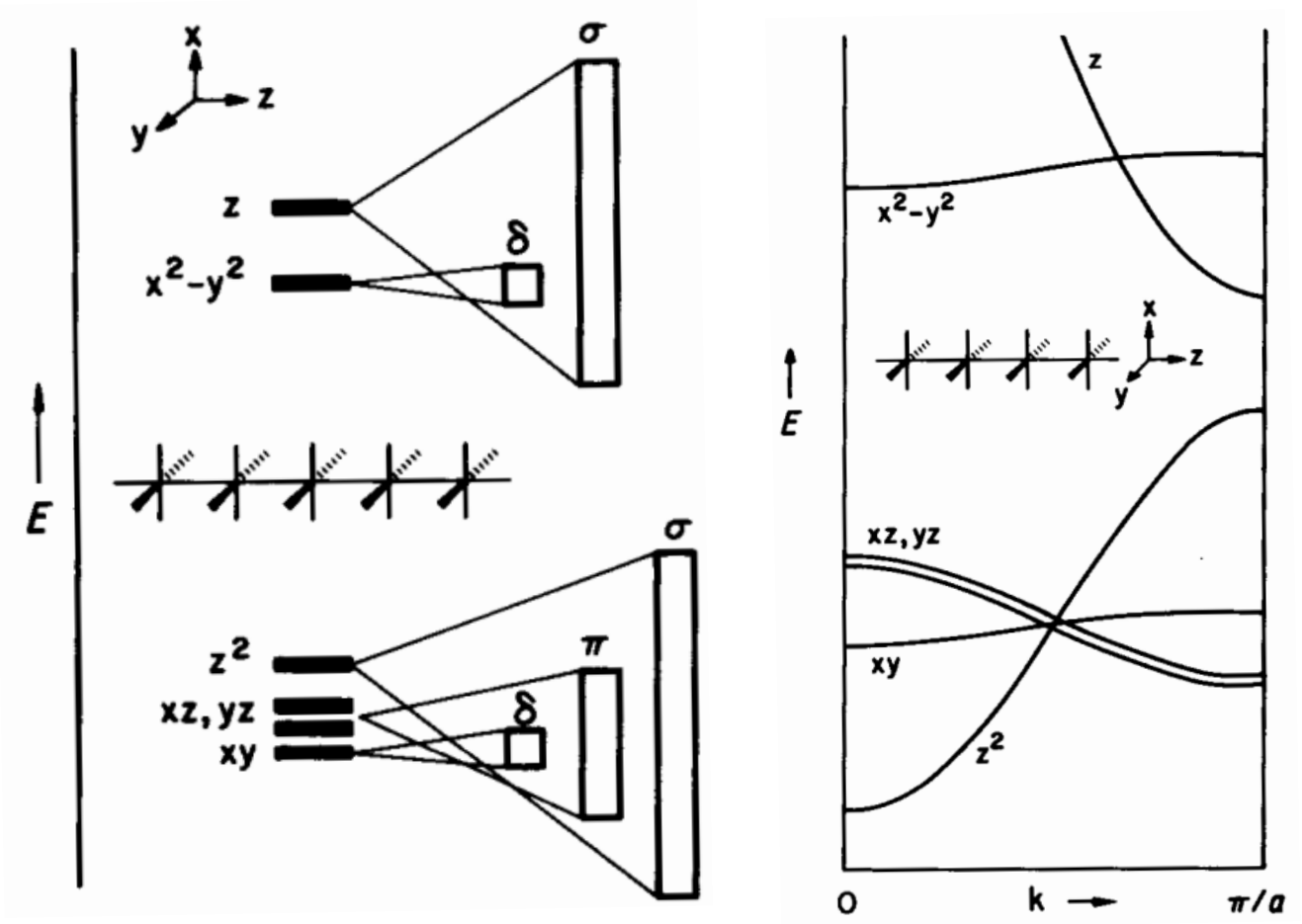
(Image: Hoffmann, *Solids and Surfaces*, p. 7 and p. 9)

“Band Width”



(Image: Hoffmann, *Solids and Surfaces*, Fig. 1)

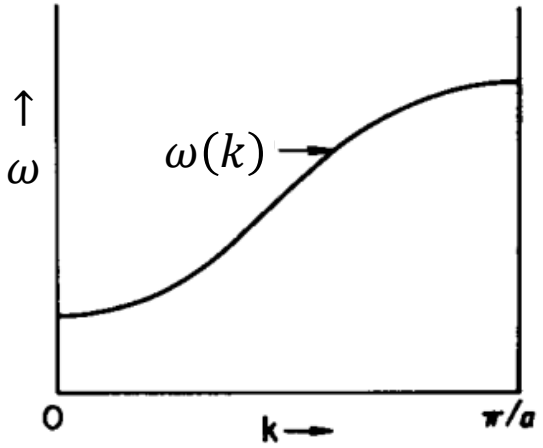
Band Structure



(Image: Hoffmann, ***Solids and Surfaces***, p. 13)

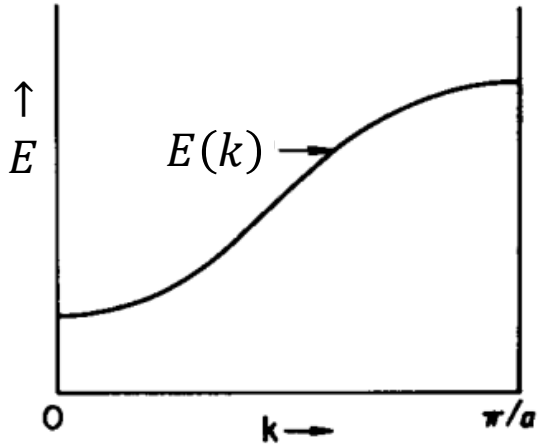
Dispersion Relations

Phonon



modes

Electron

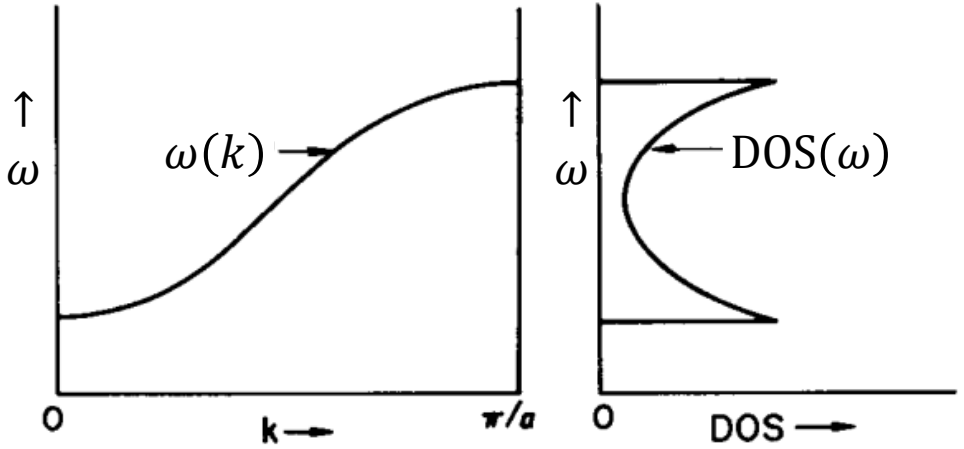


states

(Image: Hoffmann, ***Solids and Surfaces***, p. 27)

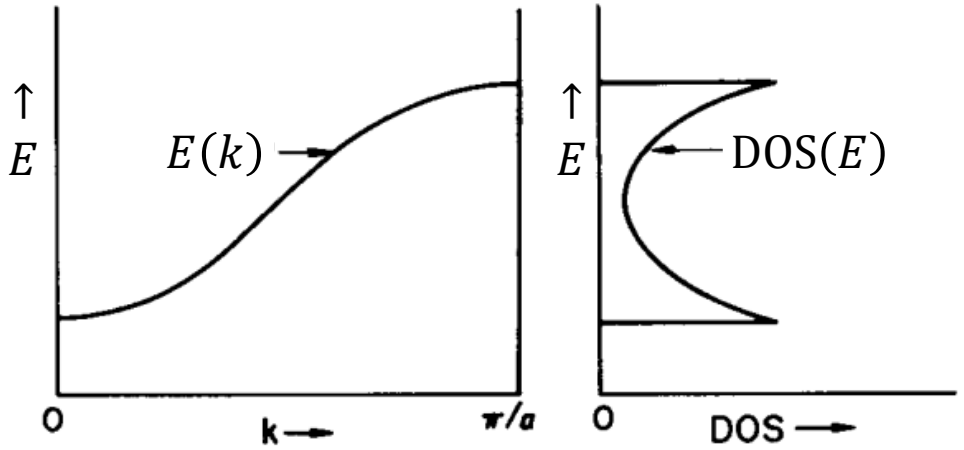
DOS (Density of States)

Phonon



modes

Electron

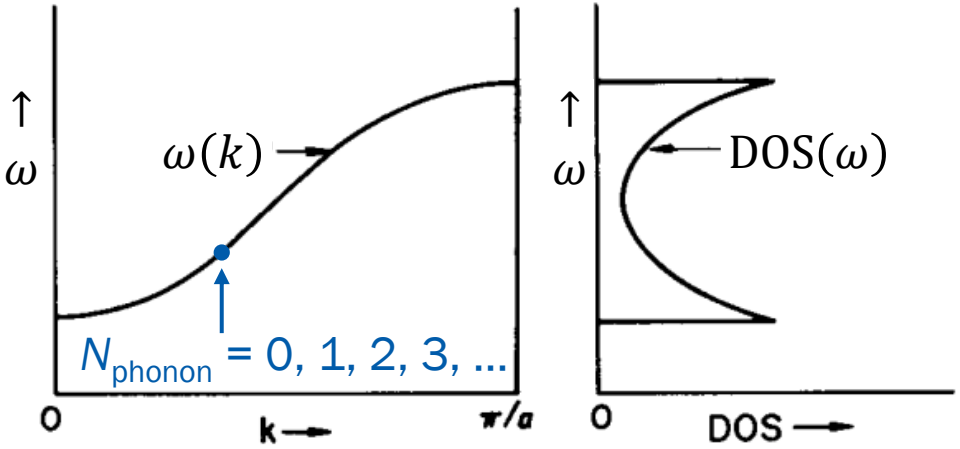


states

(Image: Hoffmann, ***Solids and Surfaces***, p. 27)

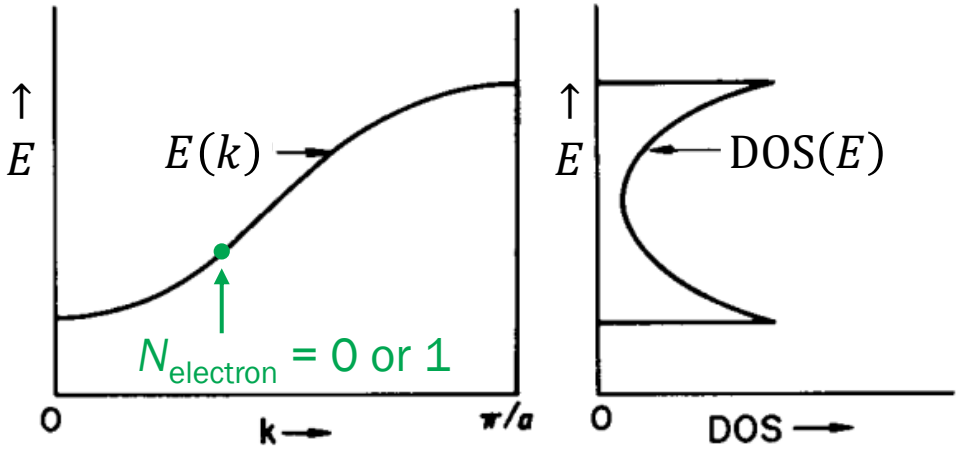
How to Fill States

Phonon



modes

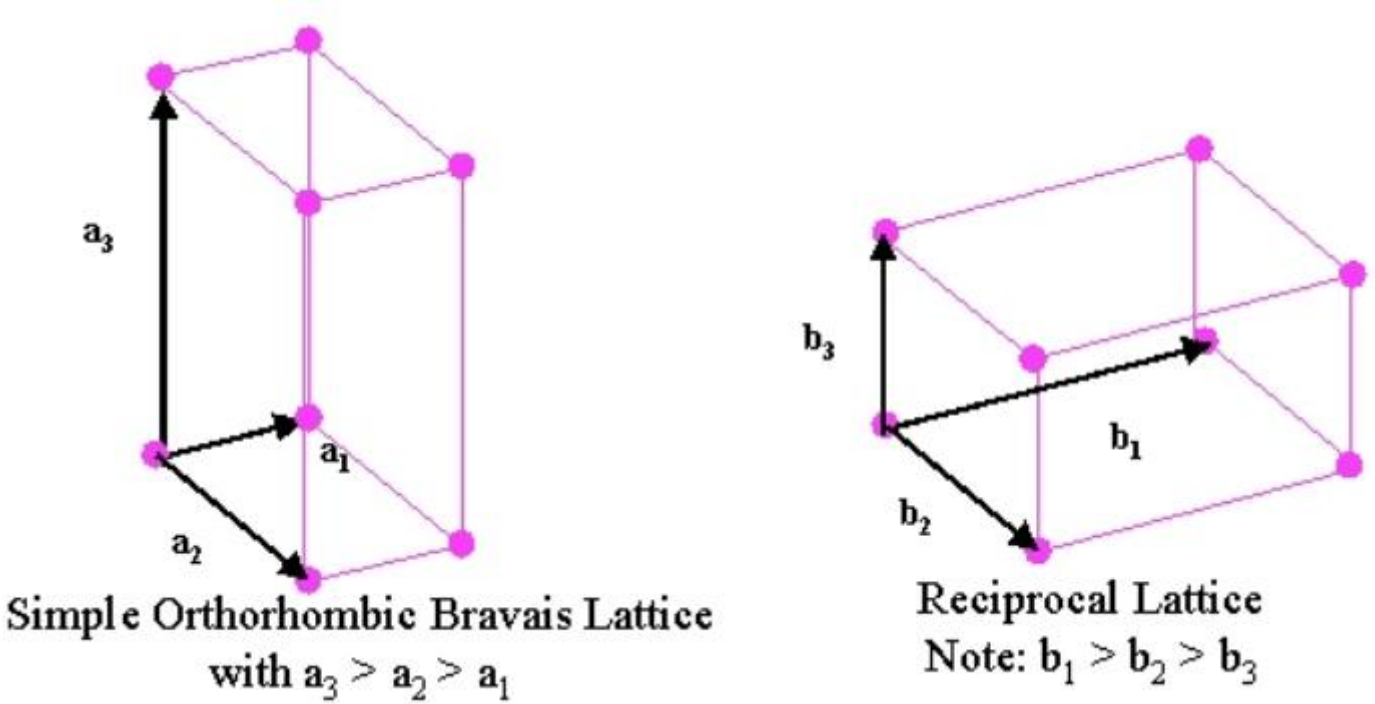
Electron



states

(Image: Hoffmann, ***Solids and Surfaces***, p. 27)

3D Reciprocal Lattice



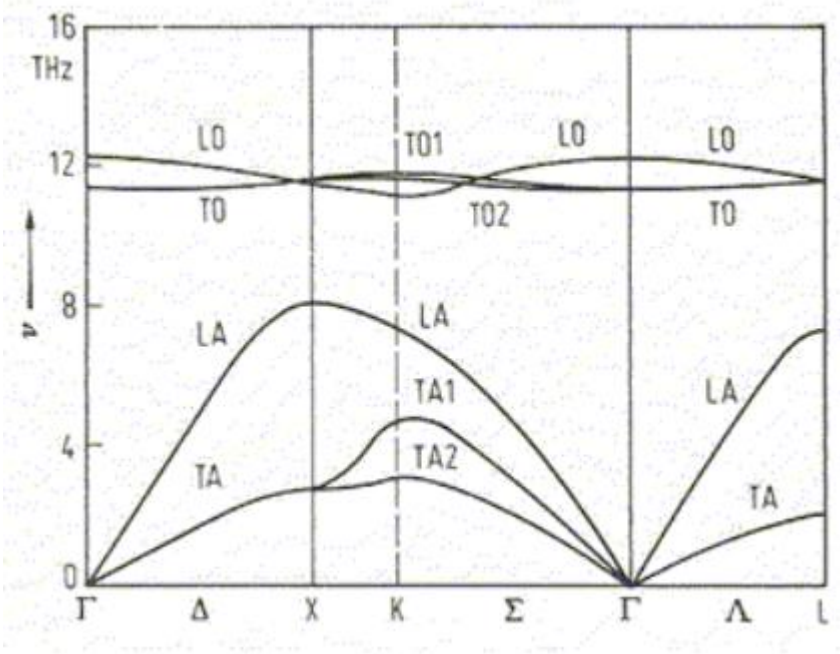
Simple Orthorhombic Bravais Lattice with $a_3 > a_2 > a_1$

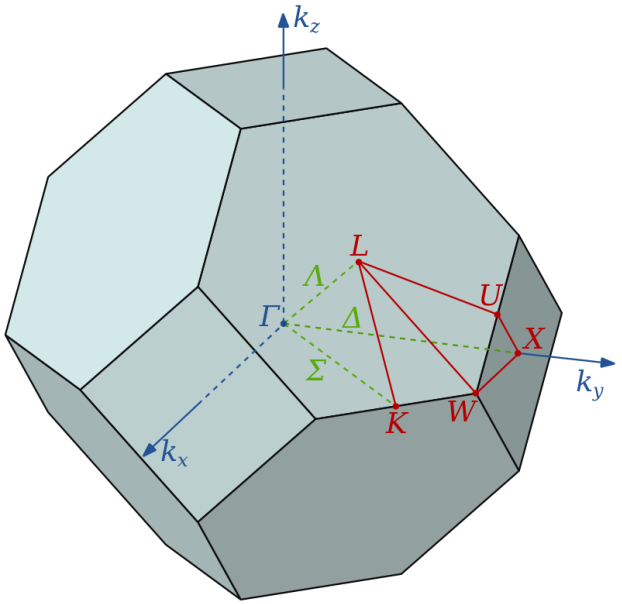
Reciprocal Lattice Note: $b_1 > b_2 > b_3$

$$\vec{a}_i \cdot \vec{b}_j = 2\pi\delta_{ij}$$

(Image: https://www.physics.smu.edu/scalise/P6338sp16/albert/chapter2.pdf)

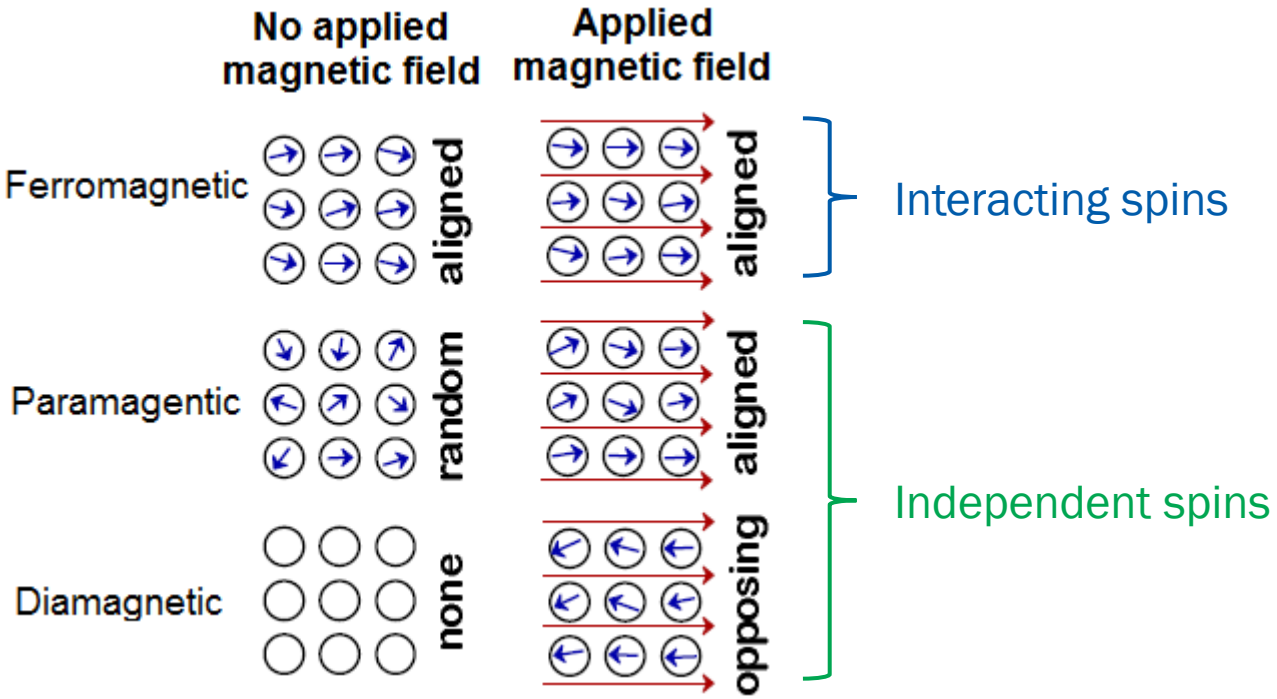
Dispersion Relation in 3D





(Images: http://lampz.tugraz.at/~hadley/ss1/problems/problems_phonons.php
https://commons.wikimedia.org/wiki/File:Brillouin_Zone_(1st,_FCC).svg)

Magnetism



(Image: https://www.researchgate.net/publication/254705371_Understanding_the_Interaction_Between_Blood_Flow_and_an_Applied_Magnetic_Field)

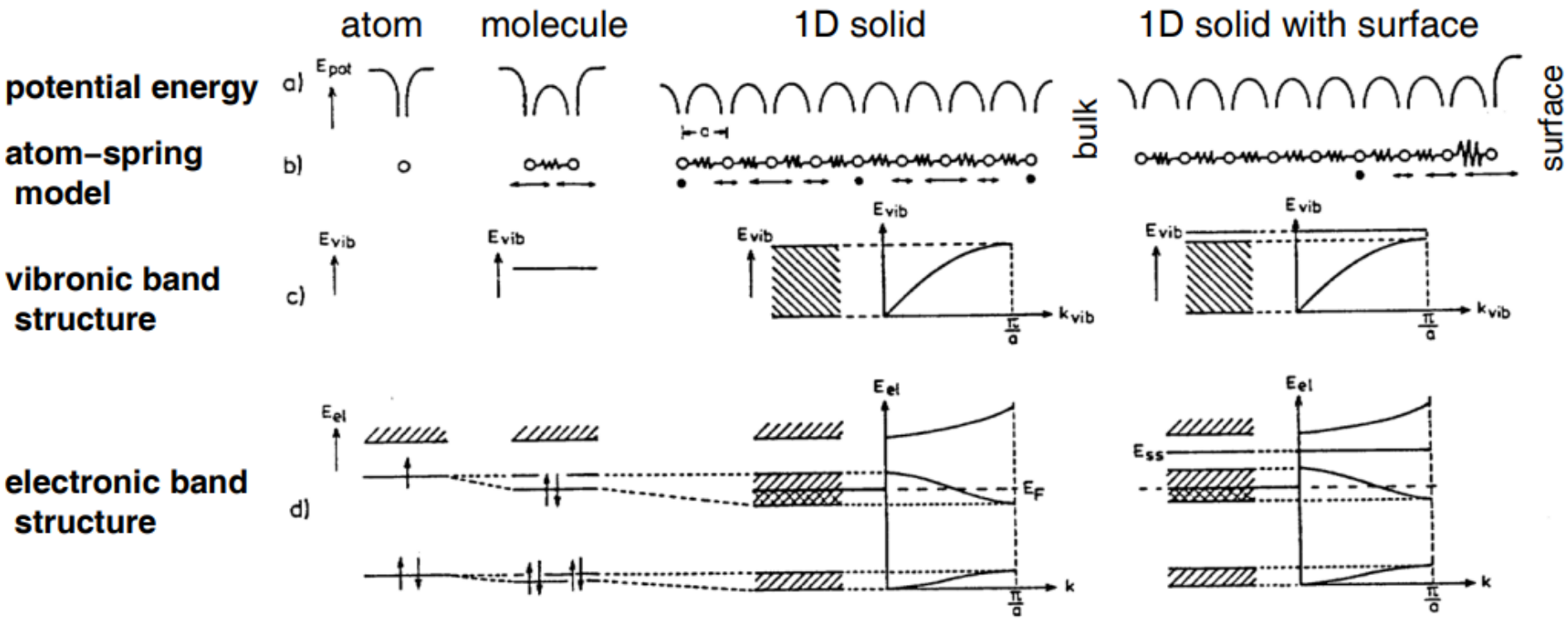
Ising Model

Consideration of neighboring spin-spin interactions

$$E = \mu_{\mathrm{B}} B \sum_i \sigma_i - J \sum_{\langle ij \rangle} \sigma_i \sigma_j$$

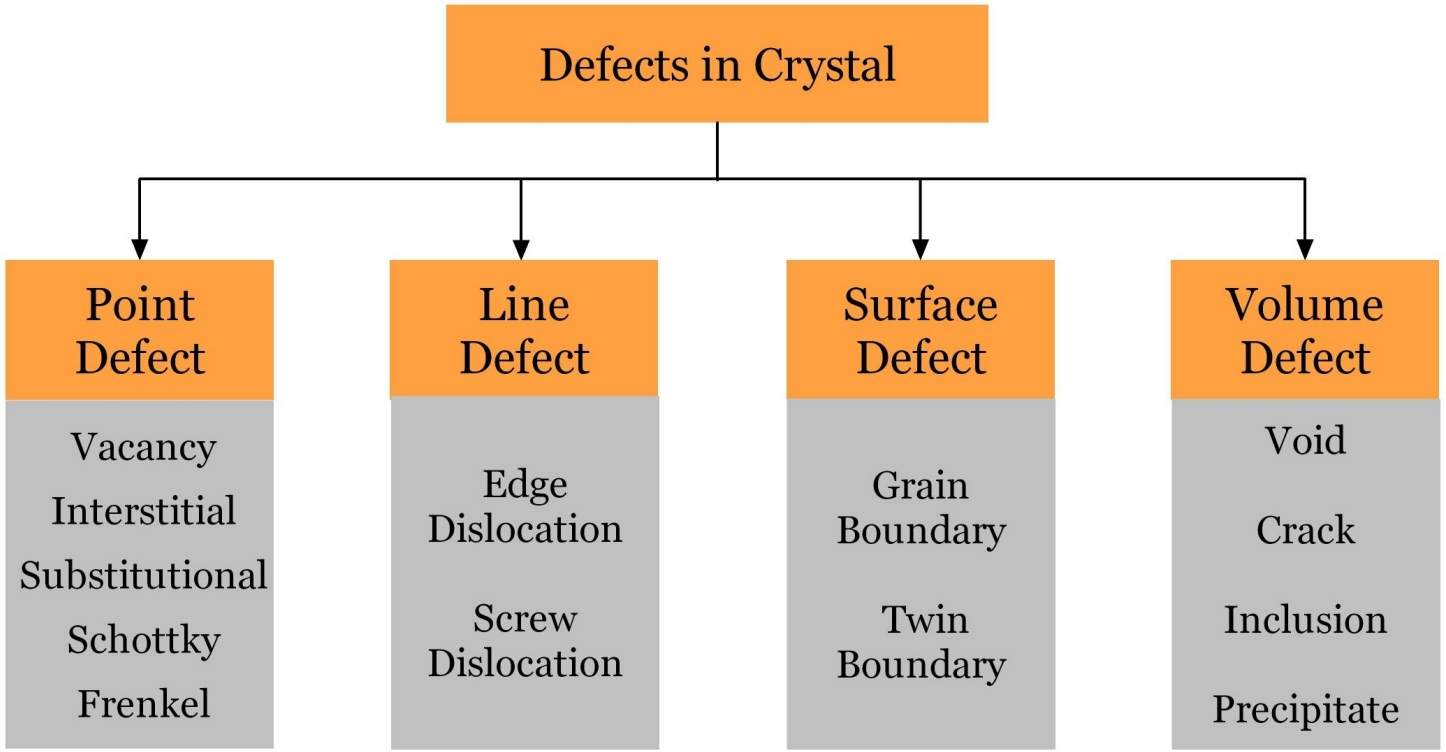
- Mean-field approach: exact for $D \geq 4$
- 1D Ising chain

Surface Effects



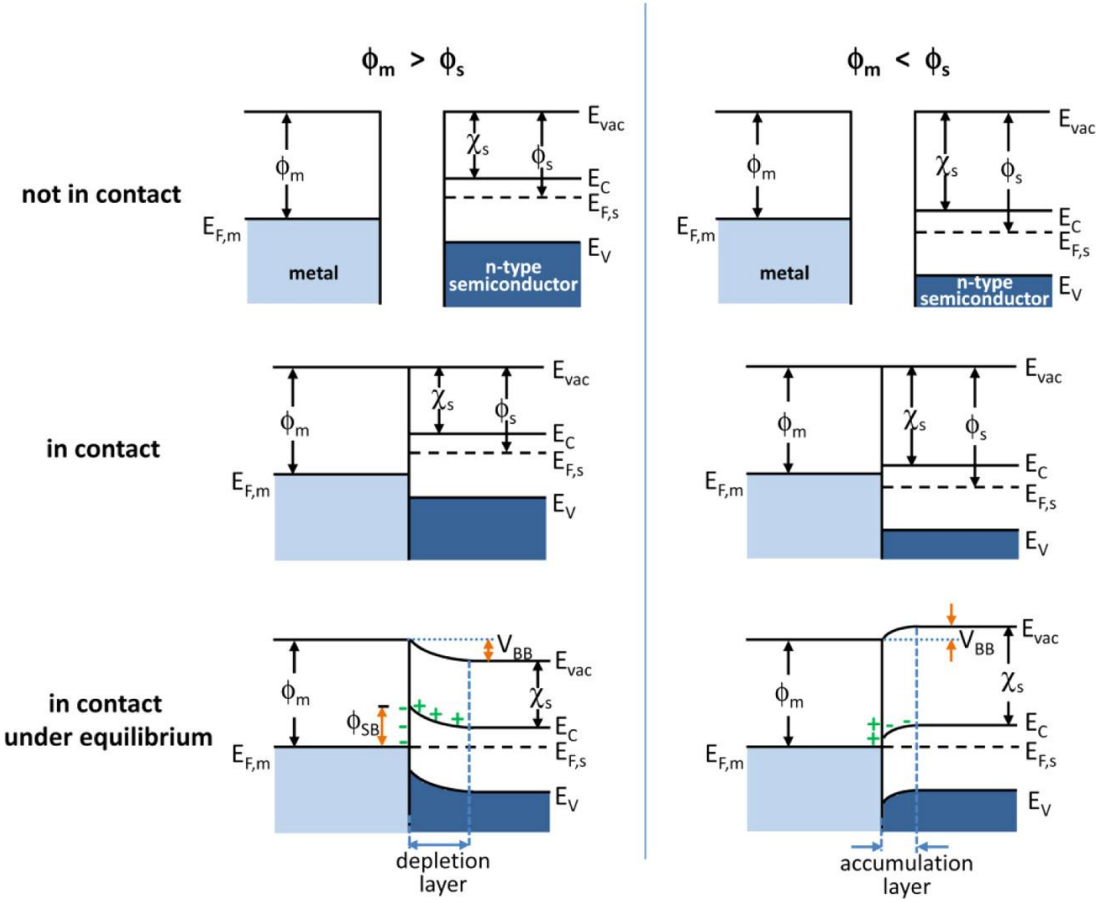
(Image: https://www.chemie-biologie.uni-siegen.de/makro/master_surf_chem/lecture_introsurfchem_1a.pdf)

Defects



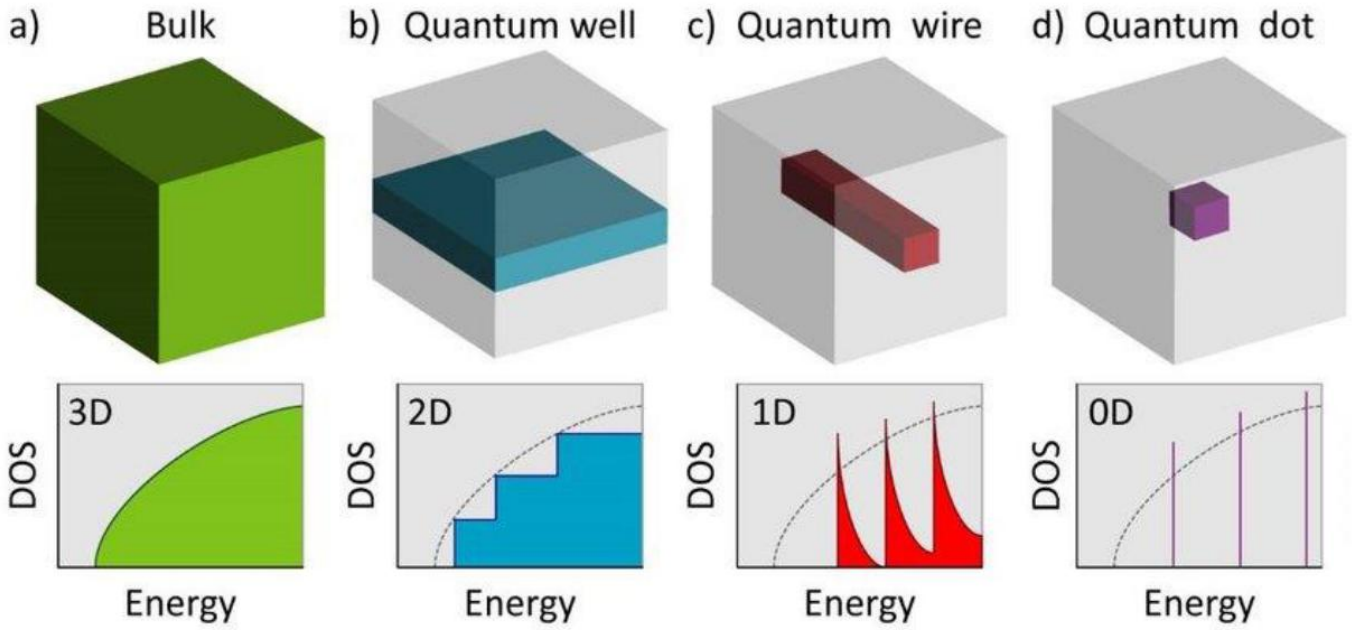
(Image: http://www.minaprem.com/materials-science/defects/vacancy-point-defect-defects-in-solid/)

Band Bending at Interface



(Image: Zhang and Yates, *Chem Rev* (2012))

Quantum Confinement on DOS



(Image: https://www.nanoinnovation2020.eu/home/PPT/Synthesis%20characteristics%20uses%20and%20detection%20of%20engineered%20nanostructured%20materials.pdf)